

Development of SneakHub: An E-Commerce Platform for Shoes

Project Submitted in Partial Fulfillment of the degree of
Master of Computer Applications



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Year 2024



July 15, 2024

To Whomsoever It May Concern

This is to certify that **Rishi Singh** is associated with Hughes Systique Pvt. Ltd. from **January 29, 2024** as a trainee under the guidance of Ms. Suman Kumar (Assistant Manager – Engineering Workforce Management).

The conduct displayed by Rishi Singh has been found good till now.

This letter is being issued to enable his to provide proof of his training to his University.

Yours sincerely

for Hughes Systique (P) Limited

Sucheta Chauhan

Manager – HR

CERTIFICATE

This is to certify that the project work entitled "SneakHub" has been carried out by **Rishi Singh** at **Hughes Systique Corporation (HSC)** from **29th Jan' 24** to **31st July' 24** under the supervision of **Ms. Suman Kumar** and **Prof. Mantosh Biswas**.

This work has been carried out for the partial fulfillment of the requirements of MCA degree in the Department of Computer Science, University of Delhi.

This project has not been submitted anywhere for the purpose of any other degree or diploma.

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Prof. Naveen Kumar



This acknowledges that

Rishi Singh

has successfully completed all the requirements to be recognized as a

**Google Cloud Certified
Cloud Digital Leader**

Series ID: 71948

Issued Date: May 01, 2024

Expiration Date: May 01, 2027

ID: 4f5dd5a1cca4421e81e831b542d83743

Certified As: Rishi Singh

A handwritten signature in black ink, appearing to read "Thomas Kurian".

Thomas Kurian
CEO, Google Cloud



Acknowledgement

I would like to express my sincere gratitude to Hughes Systique Corporation for providing me with the opportunity to intern at their esteemed organization. The experience has been invaluable, allowing me to develop my skills and gain practical knowledge in the field of software development.

I am deeply thankful to my supervisor, Ms. Suman Kumar, for their guidance, support, and constructive feedback throughout the internship. Their expertise and insights have greatly contributed to my learning and professional growth.

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Thank you all for making this internship a memorable and rewarding experience.

Sincerely,

Rishi Singh

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1. Introduction

1.1 About Hughes Systique Corporation

Hughes Systique Corporation (HSC) is a leading global provider of technology solutions and services, specializing in the development and deployment of advanced software and system integration.

Established as a prominent player in the technology consulting sector, HSC offers a wide range of services including software engineering, application development, and IT consulting. The company focuses on delivering innovative solutions that drive business efficiency and growth, leveraging cutting-edge technologies to address complex challenges across various industries.

With headquarters in Rockville, Maryland, USA, HSC operates multiple offices globally, including a major development center in Gurgaon, India. The Gurgaon office serves as a crucial hub for the company's operations, housing a significant portion of its engineering and development teams. This facility is equipped with state-of-the-art infrastructure and resources, fostering a collaborative environment where talented professionals work on groundbreaking projects.

1.2 Project Background

The SneakHub project was conceived and developed as an independent initiative during my internship at Hughes Systique Corporation. The project aimed to demonstrate my ability to create a specialized e-commerce platform dedicated to footwear. Recognizing the gap in the market for a dedicated shoe shopping experience, I undertook the challenge to build a user-friendly and efficient platform.

Utilizing advanced technologies such as Spring Boot for backend development, React for the frontend, MySQL for database management, and Docker for containerization, I aimed to showcase my technical skills and understanding of the complete software development lifecycle.

1.3 Objective of Project

The primary objective of the SneakHub project was to design and implement a comprehensive e-commerce platform specifically tailored for shoe shopping. Key objectives included:

- Demonstrating my ability to develop a responsive and attractive user interface using React.
- Showcasing backend development skills with Spring Boot to create efficient and scalable services.
- Managing data effectively using MySQL.
- Employing Docker for seamless deployment and scalability.
- Incorporating essential e-commerce functionalities such as product filtering, cart management, and a smooth checkout process to enhance the user experience.

1.4 Problem Statement

General e-commerce platforms often fail to provide a specialized and optimized shopping experience for footwear enthusiasts. Users encounter a cluttered interface, irrelevant product recommendations, and inefficient search capabilities. This results in a frustrating shopping experience, making it difficult to find and purchase desired footwear. The SneakHub project aims to address these challenges by creating a dedicated platform with advanced filtering options, streamlined cart management, and an intuitive checkout process. This project not only fills the gap in the market but also serves as a testament to my ability to develop and implement a full-fledged e-commerce solution during my internship at Hughes Systique Corporation.

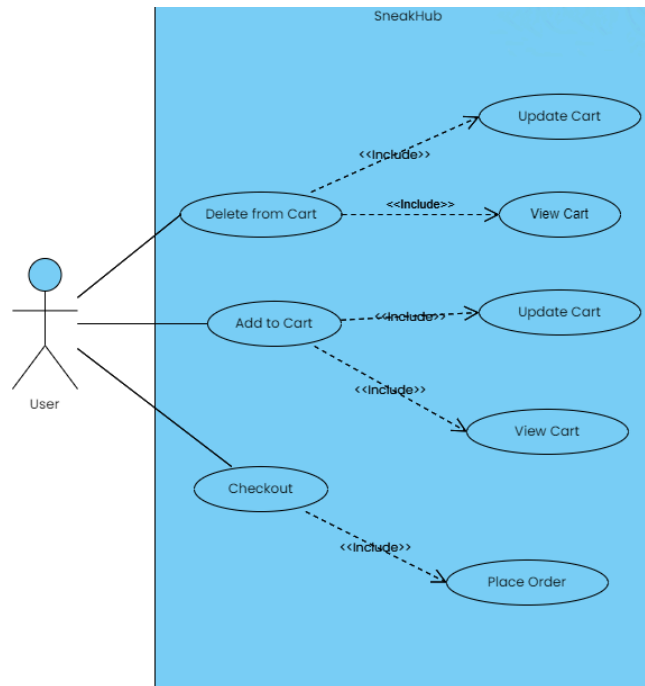
2. Proposed Solution

2.1 Process Model

The SneakHub project follows an iterative and incremental development process, aligning with Agile methodology. This approach ensures continuous integration and delivery, allowing for regular feedback and improvements. The process model includes planning, designing, implementing, testing, and reviewing each component of the system iteratively to enhance functionality and user experience progressively.

2.2 Flow Chart

Figure 1: This flow chart illustrates the various user interactions within the SneakHub application, including adding products to the cart, updating quantities, viewing the cart, and checking out.



Workflow

The workflow for SneakHub encompasses the following key processes:

1. User Registration and Login:
 - Users register by providing their details.
 - Upon registration, users can log in using their credentials.

- JWT is used for authentication and session management.

2. Product Browsing and Filtering:

- Users browse through products.
- Advanced filtering options allow users to search for products based on categories and gender.

3. Adding Products to Cart:

- Users can add products to their cart.
- The system updates the cart and displays the current items in the cart.

4. Cart Management:

- Users can view their cart.
- Users can update the quantity of items or remove items from the cart.

5. Checkout and Order Placement:

- Users proceed to checkout.
- The system processes the order and updates the inventory.

2.3 Explanation of Flowchart Steps

1. User Registration and Login:

- Users fill out a registration form with their details (name, email, password, etc.).
- The system stores user details in the **User** table.
- Users log in with their credentials, authenticated against the stored data.
- Upon successful authentication, a JWT is generated and sent to the user.
- The JWT is used for subsequent requests to verify the user's identity and maintain session security.

2. Product Browsing and Filtering:

- Users browse the product catalog.
- The system retrieves product details from the **Product** table.
- Users can filter products by category and gender to find desired items.

3. Adding Products to Cart:

- Users select products to add to their cart.
- The system updates the **Cart** table with the product ID (**pid**), user ID (**userid**), quantity, price, name, and brand.
- The cart is updated and displayed to the user, showing current items.

4. Cart Management:

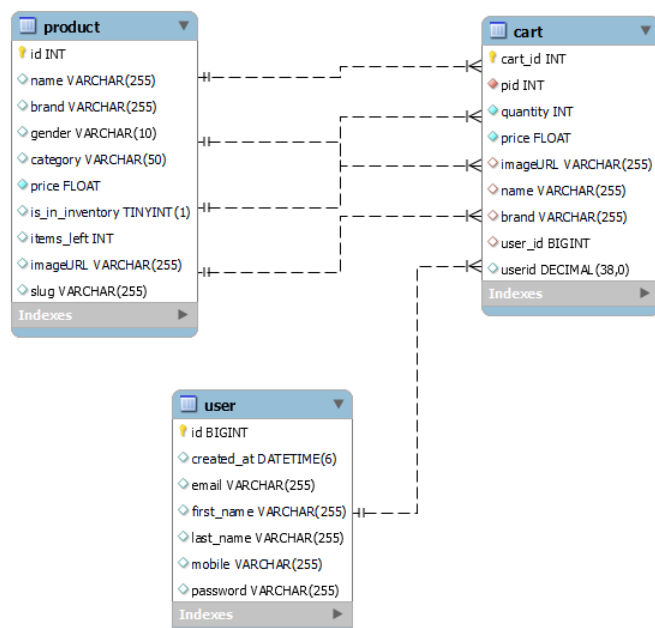
- Users view their cart contents.
- The **Cart** table is queried to retrieve current cart items.
- Users can update the quantity of items, triggering the **updateQuantity** function, which adjusts the **quantity** field in the **Cart** table.
- Users can remove items from the cart using the **deleteFromCart** function, which deletes the item entry from the **Cart** table.

5. Checkout and Order Placement:

- Users proceed to checkout and place their order.
- The system processes the order, updates inventory levels, and confirms the purchase.
- Order details are finalized, and the user receives a confirmation of their purchase.

2.4 SQL Schema Explanation:

Figure 2: The schema shows the relationships between the three main tables: Product, Cart, and User.



1. Product Table:

- Fields: `id`, `name`, `category`, `gender`, `price`.
- Stores product information available on SneakHub.

2. Cart Table:

- Fields: `cart_id`, `userid`, `pid`, `quantity`, `price`, `name`, `brand`.
- Manages user cart items, including updates and deletions.

3. User Table:

- Fields: `id`, `name`, `email`, `password`, `firstName`, `lastName`.
- Stores user details and credentials for authentication and user management.

Authentication with JWT:

- Upon successful login, a JWT is generated and provided to the user.
- The JWT is included in the header of subsequent requests for secure authentication.
- The server verifies the JWT to authenticate users and maintain session security.

3. Implementation Approach

3.1 Languages Used

The SneakHub project utilizes a combination of languages to ensure a robust and efficient system:

- Java: Used for backend development with Spring Boot, providing a strong foundation for server-side logic and operations.
- JavaScript: Employed in the frontend development using React, enabling dynamic and responsive user interfaces.
- SQL: Utilized for database interactions with MySQL, ensuring efficient data management and retrieval.

3.2 Database Integration

The database integration for SneakHub is handled through MySQL, chosen for its reliability and performance. The schema includes three primary tables:

- Product Table:
 - Fields: `id`, `name`, `category`, `gender`, `price`
 - Stores information about the products available on the platform.
- Cart Table:
 - Fields: `cart_id`, `userid`, `pid`, `quantity`, `price`, `name`, `brand`
 - Manages the items users add to their carts, tracking quantities and product details.
- User Table:
 - Fields: `id`, `name`, `email`, `password`, `firstName`, `lastName`
 - Stores user information and credentials for authentication and user management.

Integration with MySQL is achieved using Spring Data JPA and Hibernate, simplifying data access and manipulation through a repository-based approach. This ensures efficient querying and interaction with the database.

3.3 Tools Used

The following tools were employed in the development of SneakHub to streamline processes and enhance functionality:

- Spring Boot: Facilitates the creation of a robust and scalable backend, providing various features like dependency injection, security, and data access.
- Hibernate: Used as the ORM (Object-Relational Mapping) framework within Spring Boot, allowing seamless interaction with the database.
- Spring MVC: Implements the Model-View-Controller pattern, separating concerns and enabling a clear structure for request handling and view rendering.
- React: Enables the development of a dynamic and responsive frontend, offering a seamless user experience.
- MySQL: Chosen for its efficiency and reliability in managing the application's database.
- Docker: Used for containerization, ensuring consistent environments across development, testing, and production.
- JWT (JSON Web Tokens): Implements secure authentication, ensuring safe and efficient user login and session management.

4. Implementation Plan in Detail

4.1 Backend Development with Spring Boot and Hibernate

- Setting Up the Project:
 - Initialize a Spring Boot project with dependencies for Spring Data JPA, Hibernate, Spring Security, and JWT for authentication.
 - Configure application properties to connect to the MySQL database.
- Entity and Repository Creation:
 - Create entity classes for **Product**, **Cart**, and **User**, mapping them to their respective tables in the database.
 - Implement JPA repositories for each entity to handle CRUD operations.
- Service Layer:
 - Develop service classes to encapsulate business logic, such as managing products, handling cart operations, and processing user actions.
 - Use the service layer to interact with repositories and implement transaction management.
- JWT Authentication:
 - Implement authentication using JWT to secure endpoints.
 - Configure Spring Security to handle login and generate JWT tokens for authenticated users.
 - Secure routes to ensure only authorized users can access certain functionalities.

4.2 Front End Development with React

- Setting Up the React Project:
 - Initialize a React project using Create React App.
 - Set up routing with React Router for navigation between different pages.
- Component Structure:
 - Design and develop reusable components for the user interface, such as product listings, product details, cart, and checkout pages.
 - Implement state management using React's `useState` and `useContext` hooks to manage global state, such as user authentication status and cart contents.

- API Integration:
 - Create services to interact with the backend API, handling requests for product data, cart operations, and user authentication.
 - Use Axios or Fetch API to make HTTP requests to the backend.
- User Interface Design:
 - Ensure a responsive design using CSS frameworks like Bootstrap or Material-UI.
 - Focus on user experience, making the application intuitive and easy to navigate.

4.3 Database Schema and Management

- Database Schema Design:
 - Design the database schema to support the application's requirements.
 - Create tables for **Product**, **Cart**, and **User** with appropriate relationships and constraints.
- Data Access Layer:
 - Implement DAO (Data Access Object) patterns using Spring Data JPA to abstract and encapsulate all access to the data source.
 - Ensure efficient querying and handling of data operations.

4.4 Core Features Implementation

- Product Management:
 - Develop frontend components to display products, filter by category and gender, and view product details.
- Cart Operations:
 - Implement backend services to handle cart operations, including adding products to the cart, updating quantities, and removing items.
 - Develop frontend components to manage the cart, display cart contents, and update quantities.
- Checkout Process:
 - Implement the checkout process, ensuring secure and accurate order placement.
 - Handle backend logic for processing orders and updating inventory.
 - Develop frontend components to guide the user through the checkout process and confirm orders.

4.5 Scope and Limitations

Scope:

- User Management: Implement functionalities for user registration, login, authentication using JWT, and user profile management.
- Product Catalog: Develop features for browsing, searching, and filtering products by categories and gender. Display detailed product information including name, price, and brand.
- Shopping Cart: Allow users to add products to the cart, update quantities, view cart contents, and remove items from the cart.
- Order Management: Implement the checkout process, allowing users to place orders and process payments. Update inventory levels based on orders.
- Responsive Design: Ensure the application is responsive and provides a seamless user experience across different devices and screen sizes.
- Security: Secure user data and transactions through JWT authentication and proper authorization mechanisms.

Limitations:

- Payment Integration: The current implementation does not include integration with third-party payment gateways for processing payments.
- Advanced Filtering and Sorting: Limited advanced filtering and sorting options for products based on attributes such as price range, ratings, and popularity.
- User Reviews and Ratings: The system does not support user reviews and ratings for products, which could enhance user engagement and decision-making.
- Recommendation System: There is no recommendation system to suggest products based on user behavior and preferences.
- Scalability: While Docker is used for containerization, the current deployment setup might not handle very high traffic or large-scale operations without further optimization and scaling solutions.
- Multilingual Support: The application does not support multiple languages, which could limit accessibility for non-English speaking users.

- Analytics and Reporting: Limited analytical features and reporting tools for administrators to monitor sales, user activity, and other key metrics.

5. Simulations

Figure 3: The home page displaying various product categories for users to browse.

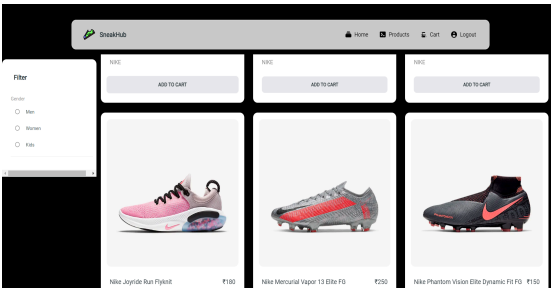
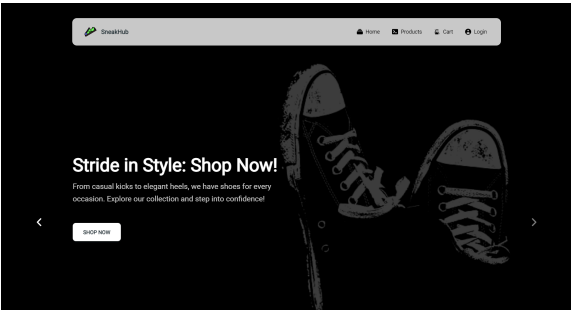


Figure 4: A detailed view of a selected product, showing its name, price, category, and other attributes.

Figure 5: The shopping cart screen where users can view, update, and remove items.

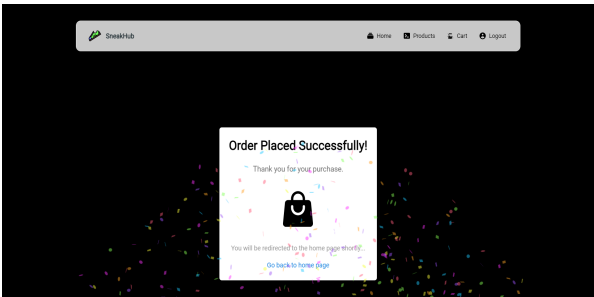
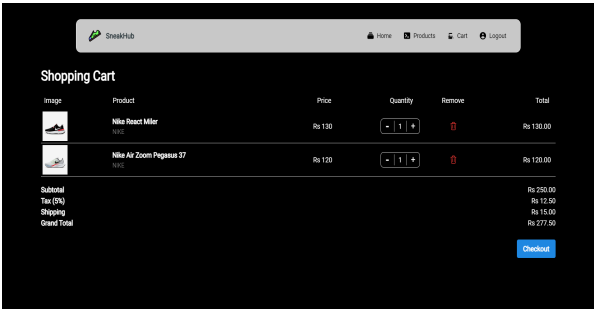


Figure 6: The checkout page where users can finalize their purchase by providing shipping details and payment information.

6. Discussion

6.1 Details of Internship

During the internship at Hughes Systique Corporation, I worked on the SneakHub project, which involved using Spring Boot, React, MySQL, Docker, and Hibernate. This experience allowed me to apply theoretical knowledge in a practical setting and gain hands-on experience in developing a full-stack e-commerce application.

6.2 Technical Skill Development

The internship significantly enhanced my technical skills, including:

- **Backend Development:** Gained proficiency in using Spring Boot and Hibernate for building scalable backend services and handling data persistence.
- **Frontend Development:** Improved my skills in React, enabling me to create dynamic and responsive user interfaces.
- **Database Management:** Developed expertise in designing and managing databases using MySQL and integrating them with the application through JPA.
- **Containerization:** Learned to use Docker for containerizing applications, which ensured consistent environments and streamlined development and deployment processes.
- **Cloud Technologies:** Acquired knowledge in cloud platforms, including obtaining the Google Cloud Platform Digital Leader certification.

6.3 Collaboration and Communication

Working as part of a team provided valuable insights into effective collaboration and communication:

- **Team Collaboration:** Coordinated with team members to ensure smooth integration of various components and adherence to project timelines.
- **Feedback and Reviews:** Participated in code reviews and incorporated feedback to enhance code quality and project outcomes.

6.5 Personal Growth

The internship contributed to my personal growth in several ways:

- Confidence: Gained confidence in applying technical skills to real-world problems and presenting solutions effectively.
- Time Management: Improved my ability to manage time and prioritize tasks, balancing multiple responsibilities and deadlines.
- Professionalism: Developed a professional attitude towards work, including attention to detail, responsibility, and commitment to quality.

Overall, the internship was a rewarding experience that provided both technical and personal development opportunities, preparing me for future challenges in the field of software development.

7. Conclusion

7.1 Summary of Experience

My internship at Hughes Systique Corporation provided an invaluable opportunity to work on the SneakHub project, an e-commerce platform for shoes. This experience allowed me to apply academic knowledge in a practical setting, develop technical skills across various technologies, and understand the intricacies of full-stack development. From backend services using Spring Boot and Hibernate to frontend development with React and database management with MySQL, the project was a comprehensive learning experience. Additionally, I gained exposure to containerization with Docker and enhanced my problem-solving abilities through real-world challenges.

7.2 Contributions

During my internship, I made several key contributions:

- **Developed Core Features:** Implemented essential functionalities for the SneakHub application, including user management, product catalog, shopping cart operations, and the checkout process.
- **Technical Expertise:** Utilized Spring Boot, Hibernate, React, MySQL, and Docker to build a robust and scalable application.
- **Enhanced Security:** Implemented JWT authentication to ensure secure user sessions and data protection.
- **Problem Resolution:** Addressed and resolved technical issues, optimizing the performance and reliability of the application.
- **Training and Certification:** Completed training in relevant technologies and earned certifications, such as the Google Cloud Platform Digital Leader, to further solidify my expertise.

7.3 Future Aspects

Looking ahead, there are several areas where the SneakHub project can be further developed and enhanced:

- **Payment Integration:** Incorporate third-party payment gateways to streamline the checkout process and improve user experience.
- **Advanced Filtering and Sorting:** Implement more sophisticated filtering and sorting options for products to enhance user navigation and search capabilities.
- **User Reviews and Ratings:** Add features for users to leave reviews and ratings on products, fostering user engagement and trust.
- **Recommendation System:** Develop a recommendation system to suggest products based on user behavior and preferences, increasing user retention and sales.
- **Scalability:** Optimize the application to handle higher traffic and larger datasets, ensuring smooth performance under increased demand.
- **Multilingual Support:** Implement support for multiple languages to make the application accessible to a wider audience.
- **Analytics and Reporting:** Integrate analytical tools and reporting features to provide insights into user behavior, sales trends, and inventory management.

8. References

1. **Spring Boot Documentation:** Comprehensive guide and reference for using Spring Boot to build robust backend applications. [Spring Boot Official Documentation](#)
2. **React Documentation:** Official documentation for React, providing in-depth tutorials and API reference. [React Official Documentation](#)
3. **MySQL Documentation:** Detailed documentation for MySQL, covering installation, configuration, and advanced usage. [MySQL Official Documentation](#)
4. **Docker Documentation:** Guides and references for using Docker to containerize applications. [Docker Official Documentation](#)
5. **Google Cloud Platform Documentation:** Resources and guides for using Google Cloud services. [Google Cloud Platform Digital Leader Learning Path](#)
6. **Axios Documentation:** Documentation for Axios, a promise-based HTTP client for making API requests. [Axios GitHub Repository](#)
7. **Tailwind CSS Documentation:** Guide and reference for using Tailwind CSS, a utility-first CSS framework for building custom designs. [Tailwind CSS Official Documentation](#)
8. **Material-UI Documentation:** Reference for using Material-UI, a popular React UI framework. [Material-UI Official Documentation](#)